

Assignment

Web Development using Python and Django

Modules Covered: Python Fundamentals, Databases with SQL, and Web Client Programming

Python Fundamentals

Explain the concepts of *sequence* and *iteration* in Python with suitable examples. Write a Python program that uses a `for` loop and a `while` loop to demonstrate iteration over a list of numbers, printing only the even numbers.

Q1.

Functions and File Operations

Write a Python program with a user-defined function `word_count(filename)` that reads a text file and returns the total number of words in it. Demonstrate file opening, reading, and closing operations, and explain the use of the `with` statement for file handling.

Q2.

Python Libraries

What is a Python library/module? Differentiate between the `import module`, `from module import name`, and `import module as alias` styles of importing, with an example for each.

Q3.

Database Fundamentals and ER Diagrams

Design an ER diagram for a *Library Management System* consisting of entities **Book**, **Member**, and **Issue**. Clearly show attributes, primary keys, and relationships (with cardinality) between the entities.

Q4.

Data Definition and Constraints

Write SQL DDL statements to create a table **Student** with fields `roll_no` (Primary Key), `name`, `email` (Unique), `dept`, and `marks` (with a `CHECK` constraint that marks must be between 0 and 100). Also write an `ALTER TABLE` statement to add a new column `phone`.

Q5.

SQL Query Categories

Differentiate between DDL, DML, DQL, DCL, and TCL commands in SQL. Give at least two example commands under each category.

Q6.

Normalization

Q7.

What is database normalization? Explain 1NF, 2NF, and 3NF with a suitable un-normalized table example, showing how it is progressively normalized to 3NF.

SQL Joins

Q8.

Consider two tables `Employee(emp_id, name, dept_id)` and `Department(dept_id, dept_name)`. Write SQL queries to demonstrate: (a) INNER JOIN, (b) LEFT JOIN, (c) RIGHT JOIN, and (d) FULL JOIN between these tables, and briefly explain the result each query produces.

Connecting to Databases using Python

Q9.

Write a Python program using the `sqlite3` module to (a) connect to a database `college.db`, (b) create a table `Student(roll_no, name, marks)`, (c) insert three records, and (d) fetch and display all records using `SELECT`.

Web Client Programming and Scraping

Q10.

Explain the basic structure of a web page (HTML, CSS, JavaScript). Write a Python program using the `requests` module and `BeautifulSoup` to download a web page, parse its HTML, extract all hyperlinks (`<a>` tags), and save the page content to the hard drive. Briefly explain the difference between a *screen scraper* and a *web crawler*.